

AI Grand Prix Physical Qualifier Technical Specification

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1. Document Control

1.1 Revision History

Issue	Date	Author	Summary of Changes
00.01	2026-08-18	NT	First release

1.2 Audience

This specification is addressed to competition participants (“Teams”). It defines the technical interface and operational requirements required to develop autonomous control software for the physical qualifier AI Drone Race.

2. Purpose and Scope

2.1 Purpose

This document provides contestants with essential information regarding the physical qualifier event, the racing environment, and the provided drone and compute hardware. Additionally, it details the software stack specifications.

2.2 Scope

- Rules
- On site program
- Specs of the software stack
- Track environment definition
- Qualification run requirements

2.3 Out of Scope

- Internal simulator architecture
- Event operations
- Contractual or commercial matters

3. Physical Qualifier

3.1 Objective

Validate autonomous drone flight capabilities through a timed obstacle course, qualifying teams based on navigation precision, gate passage count, and lap performance.

3.2 Program

On site teams will have dedicated 15min slots every day to train the the software performance in the real track

Time Slots: Multiple defined 15-minute operational slots per team per day.

Timing Activation: Runs are classified and timed starting from the first timing line cross / gate trigger.

3.3 Qualification rules

Ranking Criteria: Leaderboard ranking is strictly determined based on the timing results of valid, completed runs. In order a run being valid the drone must fly without any intervention of human pilots

Run Requirements:

- To complete a run, the teams must complete 2 laps of the track.

Qualification Requirements & Scoring

- **Qualifying Window:** Only score times recorded during the final two days of the competition count toward qualification.
- **Primary Scoring:**
 - **Complete Runs:** Competitors who successfully clear all gates are ranked primarily by the fastest total run time.

- **Incomplete Runs:** Competitors who do not complete the run are ranked behind complete runs based on the highest number of successfully passed gates.
- **Tie-Breaker:** Ties between competitors with the same number of passed gates are resolved by the fastest elapsed time to the last validly crossed gate.

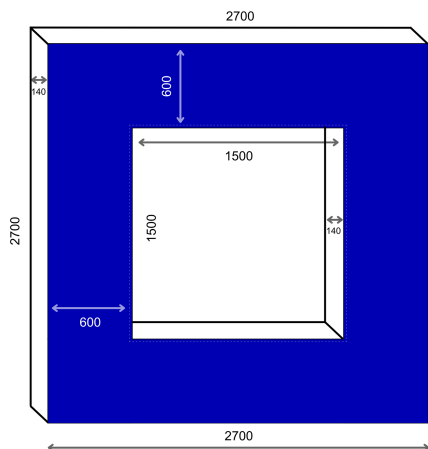
3.4 Track

- **Footprint:** 60m x 21m
- The track consists of multiple gate gates including a double gate

3.5 Gates

Gate boundaries

- Width: 2700mm
- Height: 2700mm
- Depth: 260mm



Gate inner square boundaries:

- Width: 1500mm
- Height: 1500mm
- Depth: 260mm

3.6 Drone

- **Platform:** Archer B2 Frame
- **Flight Controller (FC):** Running Betaflight firmware.
- **Inter-Board Protocol:** Serial UART protocol for high-speed telemetry and command communication with the AI compute module.

3.7 Compute Unit

Hardware:

- **System-on-Module:** NVIDIA Jetson Orin NX (16 GB)
- **Carrier / Breakout Board:** Seeed Studio A603 Carrier Board

Software Stack:

- **Base SDK:** NVIDIA JetPack 6.2
- **Provided Software:** Demo applications for camera sensor bring-up and FC-to-AI board serial communication.

3.8 Sensors

- **Vision System:** Arducam 12.3 MP HQ Camera Module with M12 Lens Mount
- IMU unit